Aim: To install and Implement typescript programming for developing and maintaining scalable and maintainable JavaScript application.

Theory:

Q1. What is TypeScript?

TypeScript is an open-source programming language developed by Microsoft that builds on top of JavaScript by adding static type definitions.

TypeScript is a programming language developed by Microsoft. It is basically JavaScript + extra features

Main difference:

- JavaScript → errors appear while running
- TypeScript → errors appear before running (during compilation)

"TypeScript helps us find mistakes early and write clean, organized code."

Q2. Why Do We Use TypeScript?

Advantages:

- Static Typing → tells what type of data is used (number, string, etc.)
- Less Errors → mistakes caught before execution
- Better Code Structure
- Supports OOP concepts (class, object, constructor)
- Converts to JavaScript → runs everywhere

Q3. List down the Software Requirements for this experiment.

- Windows/Linus/Mac Operating System
- Visual Studio Code
- Node.js
- Typescript Compiler

Q4. Write a program to print 'Hello World' via Typescript.

Code:

```
console.log("Hello World");
```

Ouput:

```
spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/WEBX/Experiments/EXP3 (master)
$ node hello.ts
Hello World
```

Q5. Write a program to perform basic arithmetic operations in Typescript.

Code:

```
import * as readlineSync from "readline-sync";

const num1Str = readlineSync.question("pehla number daalo bhai: ");
const num1 = parseFloat(num1Str);

const num2Str = readlineSync.question("doosra number daalo pleej: ");
const num2 = parseFloat(num2Str);

const operator = readlineSync.question("kya karna hai in dono number ka (+, -, *, /, %,
**): ", {
    limit: ["+", "-", "*", "/", "%", "**"],
    limitMessage: "sahi operator daalo pleej (+, -, *, /, %, **)",
});

if (isNaN(num1) || isNaN(num2)) {
    console.error("Error: number daalona pleej");
}
let result: number;
try {
    switch (operator) {
        case "+":
            result = num1 + num2;
            break;
        case "-":
            result = num1 - num2;
            break;
        case "*":
            result = num1 * num2;
            break;
        case "/":
            if (num2 === 0) {
                throw new Error("zero se divide nahi hota hai bro");
            }
            result = num1 / num2;
            break;
        case "%":
            result = num1 % num2;
            break;
        case "**":
            result = Math.pow(num1, num2);
            break;
        default:
            throw new Error(`galat operator: ${operator}`);
    }

    console.log(`\n${num1} ${operator} ${num2} = ${result}`);
} catch (error) {
    if (error instanceof Error) {
        console.error(`Error: ${error.message}`);
    }
}
```

Ouput:

```
spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/WEBX/Experiments/EXP3 (master)
$ node calc.ts
pehla number daalo bhai: 71
doosra number daalo pleej: 5
kya karna hai in dono number ka (+, -, *, /, %, **): -

71 - 5 = 66

spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/WEBX/Experiments/EXP3 (master)
$ node calc.ts
pehla number daalo bhai: 60
doosra number daalo pleej: 6
kya karna hai in dono number ka (+, -, *, /, %, **): +

60 + 6 = 66

spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/WEBX/Experiments/EXP3 (master)
$ node calc.ts
pehla number daalo bhai: 11
doosra number daalo pleej: 6
kya karna hai in dono number ka (+, -, *, /, %, **): *

11 * 6 = 66

spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/WEBX/Experiments/EXP3 (master)
$ node calc.ts
pehla number daalo bhai: 660
doosra number daalo pleej: 10
kya karna hai in dono number ka (+, -, *, /, %, **): /

660 / 10 = 66

spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/WEBX/Experiments/EXP3 (master)
$ node calc.ts
pehla number daalo bhai: Modi
doosra number daalo pleej: Amit
kya karna hai in dono number ka (+, -, *, /, %, **): +
Error: number daalona pleej

NaN + NaN = NaN

spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/WEBX/Experiments/EXP3 (master)
$ node calc.ts
pehla number daalo bhai: -67
doosra number daalo pleej: +1
kya karna hai in dono number ka (+, -, *, /, %, **): +

-67 + 1 = -66
```